Weighted single-level versus multilevel: a replication of Koziol (2015)

4 November 2026

Abstract

We partly replicated the headline result of Koziol (2015). All 208 settings were run at 1,000 replications. Weighted single-level estimators tended to outperform weighted multilevel estimators, in the same direction as the dissertation. The normal intercept contrast is 0.152 against the printed 0.151, the Bernoulli threshold contrast is −0.171 against −0.170 ($\tau = -\gamma$ in $P(Y = 1) = \Phi(\gamma + \cdots)$), and the random-coefficient slope contrast is 0.034 against 0.033 (Koziol, 2015). The first two miss the 0.001 rounding rule and are mismatches. The slope contrast is within one last-place unit and is close. Of 3,200 compared appendix cells, 1,040 match the printed entry and 1,531 are within one unit in the last printed place. Coverage matches in 90 of 752 cells. Seventy-two bracketed marginal root-mean-square-error cells were not computed. The finished factorial is 208 of 208 settings. Coverage of the weighted multilevel intercept at average cluster size 20 and intraclass correlation .05 is 0.252 against the printed 0.282, the factors $G/(G-1)$ and $N/(N-1)$ move our coverage by about 0.001, and the Monte Carlo standard error of that coverage is 0.014. The source software is Mplus 7. The tolerance on a difference of two three-decimal biases is 0.001, and a positive difference means our Monte Carlo mean is larger than the printed value.

Table 1: Signed contrasts at the cells that carry the headline. Published values are the printed contrasts in Tables C.2, C.4, and E.2 of Koziol (2015). Our values are Monte Carlo means. Rows 1, 2, and 4 are in outcome units. Row 3 is in probit threshold units. Rows 5 and 6 are proportions. The tolerance on a difference of two three-decimal biases is 0.001. Magnitude uses the unrounded mean: a gap of at most 0.001 is same.

Estimand	Published	Ours	Effect magnitude	Direction	Verdict
Normal wtm minus ws intercept	0.151	0.152	larger	same direction	mismatch
Approx. minus true intercept	0.157	0.156	same	same direction	close
Bernoulli wtm minus ws threshold	−0.170	−0.171	larger	same direction	mismatch
RC slope, approx. minus true	0.033	0.034	same	same direction	close
Normal ws coverage, cluster size 5	0.946	0.945	same	same direction	close
Weighted multilevel coverage, cluster size 20	0.282	0.252	smaller	same direction	mismatch

The source software is Mplus 7. The tolerance on a difference of two three-decimal biases is 0.001. A positive difference means our Monte Carlo mean is larger than the printed value. A positive estimand means the first estimator named in that row exceeds the second. Descriptive point estimates; not equivalence.

1 Introduction

We partly replicated the headline result of Koziol (2015). Under informative sampling, weighted single-level estimators tended to outperform weighted multilevel estimators. Our normal intercept contrast

is 0.152 against the printed 0.151, on 1,000 replications. Our Bernoulli threshold contrast is −0.171 against −0.170, on 1,000 replications, with $\tau = -\gamma$ in $P(Y = 1) = \Phi(\gamma + \cdots)$. Our random-coefficient approximate-weight minus true-weight slope contrast is 0.034 against 0.033, on 1,000 replications. The finished factorial is 208 of 208 settings. At average cluster size 20 and intraclass correlation .05, weighted multilevel coverage is 0.252 against the printed 0.282. The factors $G/(G-1)$ and $N/(N-1)$ move that coverage by about 0.001, and the Monte Carlo standard error of the coverage is 0.014. The source software is Mplus 7 (Muthén and Muthén, 2012). The tolerance on a difference of two three-decimal biases is 0.001. A positive difference means our Monte Carlo mean is larger than the printed value. These comparisons are descriptive point estimates, not equivalence tests.

Natalie A. Koziol concluded that, when selection probabilities are unequal and informative, a weighted single-level estimator with sandwich standard errors recovers the intercept more accurately than a weighted multilevel pseudo-maximum likelihood estimator, and that approximate cluster weights leave more bias than the true multilevel weights (Koziol, 2015). The dissertation is *A Comparison of Population-Averaged and Cluster-Specific Approaches in the Context of Unequal Probabilities of Selection* (University of Nebraska–Lincoln, 2015), chaired by James A. Bovaird, and archived at `https://digitalcommons.unl.edu/cehsdiss/229/`. The three research questions ask how population-averaged and cluster-specific estimators behave under informative sampling, how their interval estimates cover, and whether an approximate cluster weight can stand in for the true multilevel weight. Table 1 answers those questions on the full factorial of 208 settings. A population-averaged fit is a single-level model with a sandwich variance. A cluster-specific fit is a multilevel model with a random intercept, or with a random intercept and a random slope. A sample is informative when the selection probability depends on the cluster effect that also enters the outcome, so an unweighted fit estimates a different intercept from the census model.

Five estimators appear in the tables. The labels below are hers. Unweighted multilevel (`um.ccs`) and unweighted single-level (`us.ics`) ignore the weights. Weighted single-level (`ws.ics`) uses the selection weights and a cluster sandwich (Binder, 1983; Lumley, 2004). Weighted true multilevel (`wtm.ccs`) uses the multilevel weights inside a pseudo-likelihood (Pfeffermann et al., 1998; Asparouhov, 2006; Rabe-Hesketh and Skrondal, 2006). Weighted approximate multilevel (`wam.ccs`) replaces the cluster weight by M divided by the realized number of clusters, and divides the conditional weight by that constant, the approximation of Kovačević and Rai (2003). M is the number of clusters in the population. The population-averaged estimating equation is the same class of model that geepack implements for generalized estimating equations (Højsgaard et al., 2006). The single-level fits in this note use the survey package, because the design is a probability sample with a cluster sandwich (Lumley, 2024).

2 What do the four arms show?

The question is what the normal and Bernoulli random-intercept arms, and the normal and Bernoulli random-coefficient arms, show at 1,000 replications. The answer is that the estimator order in her tables reappears on the informative cells, and that coverage is the column that misses most often.

2.1 Normal random intercept

The question is whether weighted multilevel intercept bias exceeds weighted single-level intercept bias when the outcome is normal. The answer is yes, on the primary cell and on the rest of the eight-cell grid.

The primary cell is informative sampling, intraclass correlation .05, and average cluster size 5. Koziol reported the contrast of `wtm.ccs` minus `ws.ics` intercept bias as 0.151 in Table C.2. Our contrast on

1,000 replications is 0.152. The paired Monte Carlo standard error is 0.000415. The unrounded gap is 0.001025, which is a mismatch under the 0.001 rule. If 0.151 is treated as exact, the gap is about two and a half of our standard errors. The order is unchanged: weighted multilevel bias is the larger positive number. In the same cell the approximate-weight contrast, `wam.ccs` minus `wtm.ccs`, is 0.156 against the printed 0.157, with paired standard error 0.000514. Both contrasts are in outcome units. The cell means are stored in `simulation/results/summary.csv`, and the replicate file is `simulation/results/cell_01.csv`.

Our intercept biases on that cell are 0.682 unweighted, 0.309 with approximate weights, 0.153 with true multilevel weights, and 0.001 with the weighted single-level estimator. The printed true-multilevel bias is 0.151, so the raw multilevel mean, 0.153, misses the printed thousandth by 0.002, and the contrast with the single-level estimator is a mismatch at the same rule. Weighted single-level coverage of the intercept is 0.945 against the printed 0.946 in Table E.2, which is close. Weighted multilevel coverage is 0.000 against the printed 0.001, which is close. The coverage contrast is 0.945 against 0.945. A coverage near .95 on 1,000 draws has a binomial standard error near 0.007. She reported that the weighted single-level intervals sit near the nominal rate and the weighted multilevel intervals do not. Our run shows the same split. The bias means use all 1,000 replications. Weighted multilevel coverage on this cell uses 999 replications with a finite standard error.

The other informative cells keep the same order. At intraclass correlation .25 and average cluster size 5, our true-multilevel intercept bias is 0.163 against the printed 0.164, and the weighted single-level bias is 0.003 against 0.005. Both are mismatches at the thousandth, and the multilevel bias is still the larger one. At average cluster size 20 the true-multilevel bias is 0.044 against 0.043 (intraclass correlation .05, a mismatch) and 0.047 against 0.046 (intraclass correlation .25, close). The weighted single-level bias is 0.001 and 0.002 in those two cells. Approximate weights remain worse than true weights in every informative cell. Under non-informative sampling the intercept biases stay small. Across the four non-informative cells and five estimators, her printed intercept biases lie between -0.002 and 0.001, and ours lie between -0.001 and 0.002. The largest absolute bias is 0.002 in both sources (Table C.1). One signed cell misses: true multilevel weights at intraclass correlation .25 and average cluster size 5 are 0.002 against the printed -0.002. Relative bias is undefined at a zero intercept, so the comparison is the absolute bias.

The realized primary sample has mean 967.7 clusters and 4,884 students, and the mean cluster weight is 49.5. Table 4.1 reports a mean of 964 clusters and a mean cluster weight of about 50. A difference of about four clusters on a mean near 964 is under one percent. Cluster counts are means of Poisson samples drawn with separate seeds, so exact equality is not expected. The design check supports the bias comparison. It is not itself a performance result.

2.2 Bernoulli random intercept

The question is whether the weighted multilevel threshold is more negative than the weighted single-level threshold when the outcome is Bernoulli. The answer is yes, on every informative cell, each at 1,000 replications.

The outcome model used here is $P(Y = 1) = \Phi(\gamma + x\beta + u)$. Table C.2's intercept column matches the threshold $\tau = -\gamma$, not the latent intercept γ. At average cluster size 5 and intraclass correlation .05, our threshold means are -0.714 unweighted against the printed -0.712, -0.002 weighted single-level against 0.000, -0.173 true-weight multilevel against -0.170, and -0.363 approximate-weight against -0.361. Each of those four is a mismatch at the thousandth. The contrast of true-weight multilevel minus weighted single-level is -0.171 against the printed -0.170. The paired Monte Carlo standard error on 1,000 replications is 0.000471. The unrounded gap is -0.001098, which misses the 0.001 rule and is about two and a third of our standard errors. The sign is the same, and the four estimator means

sit in the same order as Table C.2. The cell comparison is simulation/results/cell_match.csv.

At average cluster size 20 and the same intraclass correlation, the true-weight threshold mean is −0.045 against the printed −0.043, and the approximate-weight mean is −0.209 against −0.208. Both are mismatches. The unweighted threshold mean is −0.716 against −0.713. The weighted single-level mean is −0.002 against 0.000. The multilevel threshold is still the more negative of the two weighted estimators. The same order holds at intraclass correlation .25: true-weight means are −0.200 against −0.203 at cluster size 5 and −0.057 against −0.059 at cluster size 20, and both are more negative than the weighted single-level means, which sit near zero. All eight Bernoulli random-intercept cells were run at 1,000 replications.

2.3 Normal random coefficient

The question is whether approximate cluster weights bias the level-1 slope when the true multilevel weights do not. The answer is yes, on the informative normal cell at intraclass correlation .05 and average cluster size 5, and the same factorial was run for the other seven normal cells and for all eight Bernoulli random-coefficient cells.

The headline cell is a normal outcome, informative sampling, intraclass correlation .05, and average cluster size 5. Table C.4 prints the contrast of approximate-weight slope bias minus true-weight slope bias as 0.033. Our contrast on 1,000 replications is 0.034. The paired Monte Carlo standard error is 0.000409. The unrounded gap is 0.000716, which is within one last-place unit, so the contrast is close. The slope bias subtracts the common true slope stored on each row. All 1,000 replications contributed a finite slope.

The component biases on that cell, ours against the printed value, are these. The unweighted intercept is 0.684 against 0.684, a match, and the unweighted slope is 0.033 against 0.034, close. The weighted single-level intercept is 0.001 against 0.002, close, with Monte Carlo standard error 0.000941, and the slope is −0.001 against 0.002, a mismatch. The true-weight multilevel intercept is 0.173 against 0.174, close, and the slope is 0.002 against 0.003, close. The approximate-weight intercept is 0.333 against 0.333, a match, and the slope is 0.036 against 0.036, a match. The slope contrast is the result that carries this design. The cell comparison for the other random-coefficient settings, including the Bernoulli arm, is in simulation/results/cell_match.csv.

3 What did we do, and what did we take from Koziol?

The question is which parts of the numerical work are ours and which parts are taken from the dissertation. The answer is that the design, the target parameters, and the printed tables are hers, and the Monte Carlo means in this note are ours.

Koziol specified a finite population, Poisson sampling at the cluster and student levels, normal and Bernoulli outcomes, random-intercept and random-coefficient structures, intraclass correlations .05 and .25, and average cluster sizes 5 and 20. Chapter III and Appendix A give the operational weight formulas and a fragment of the R generator. Page 76 names Mplus version 7 for estimation, with `TYPE=COMPLEX` or `TWOLEVEL`, `ESTIMATOR=MLR`, `LINK=PROBIT`, `WTSCALE=CLUSTER`, and `BWTSCALE=UNSCALED` (Muthén and Muthén, 2012). Tables C.2, C.4, and E.2 supply the published biases and coverages used above. Table 4.1 supplies the published sample-size moments. The 2017 article in *Multivariate Behavioral Research* repeats the comparison (Koziol et al., 2017). Its Figshare supplement repeats tables and does not contain Mplus or R programs. The article PDF was not re-read for this note. Every published number below is marked as taken from those tables. Every Monte Carlo mean is marked as our simulation. Choices that her deposited programs would have fixed, and that we set in order to run, are marked as assumptions in the next section.

Our reconstruction is in R 4.3.3 (R Core Team, 2024), https://www.R-project.org/. The dissertation generates the populations in R with MASS. Our generator is simulation/generate.R. Weighted single-level fits and their cluster-robust standard errors use survey 4.2.1 (Lumley, 2004, 2024), https://CRAN.R-project.org/package=survey. Unweighted Gaussian multilevel fits, and the unweighted probit start, use lme4 1.1.35.1 (Bates et al., 2015), https://CRAN.R-project.org/package=lme4. The random-slope pseudo-likelihood, and a one-replicate check of the normal multilevel fit, use WeMix 4.0.3 (Bailey et al., 2023), https://CRAN.R-project.org/package=WeMix. On that production replicate, WeMix and the analytic pseudo-likelihood agree. Numerical derivatives for the pseudo-likelihood sandwich use numDeriv 2016.8-1.1 (Gilbert and Varadhan, 2019), https://CRAN.R-project.org/package=numDeriv. The Bernoulli multilevel estimator is a Gauss–Hermite pseudo-likelihood with 15 nodes, started at the unweighted `glmer` fit and optimized with `nlminb`. The run record is simulation/results/RUN.txt.

All 208 settings were run at 1,000 requested replications. That is 40 normal random-intercept settings, 40 Bernoulli random-intercept settings, 64 normal random-coefficient settings, and 64 Bernoulli random-coefficient settings. Mplus 7 was not available for a refit of the same samples, and the original input files were not found.

4 What assumptions did we supply?

The question is which choices we supplied because the estimation programs were not deposited. The answer is the seven choices in this section. A setting she states in the dissertation stays hers.

4.1 Threshold sign

The question is which Bernoulli sign we compare with Table C.2. The answer is the threshold $\tau = -\gamma$, where γ is the intercept in $P(Y = 1) = \Phi(\gamma + x\beta + u)$.

The latent intercept has the opposite sign from the column she printed. The threshold matches that column, including the unweighted mean near -0.71 and the approximate-weight mean near -0.36. This is a parameterization that matches the table. It is not a finding that the table is wrong.

4.2 Method 2 scaling

The question is how the multilevel weights are scaled before the pseudo-likelihood. The answer is the cluster scaling she names as `WTSCALE=CLUSTER`, which this reconstruction treats as Method 2, with between-cluster weights left unscaled (`BWTSCALE=UNSCALED`).

Asparouhov describes scaling choices for multilevel sampling weights, and the Mplus 7 user's guide names the option on the page she cites (Asparouhov, 2006; Muthén and Muthén, 2012; Rabe-Hesketh and Skrondal, 2006). We did not refit the grid under the other scale options. A different scale would change the weighted multilevel estimator, so this choice is part of the comparison rather than a harmless default.

4.3 Normal intervals at 1.96

The question is which critical value defines coverage. The answer is 1.96.

Coverage is the proportion of replications in which the Wald interval $\hat{\theta} \pm 1.96\,\mathrm{SE}$ contains the target. We did not use a t critical value and we did not apply a further degrees-of-freedom correction on top of the

sandwich. Table E.2 is compared with that definition. The single-level standard errors are the survey cluster sandwich. The weighted multilevel standard errors are the observed pseudo-likelihood sandwich.

4.4 Separate seeds

The question is whether our samples are her samples. The answer is no.

Her seed stream was not deposited. Each cell in this reconstruction uses its own seed sequence. Monte Carlo means are the objects compared with the printed tables. A single replicate is not expected to match a replicate she printed, and she did not print replicate-level estimates. The cluster-count gap of about four clusters is the sort of difference separate seeds produce.

4.5 Model-based standard errors in lme4

The question is where the unweighted Gaussian multilevel standard errors come from. The answer is the model-based variance from lme4, not an Mplus MLR sandwich.

The weighted fits use a sandwich, as her MLR columns do. The unweighted multilevel coverage in our normal grid is therefore not a reconstruction of `ESTIMATOR=MLR`. The headline coverage contrast uses the weighted single-level sandwich and the weighted multilevel sandwich, so this assumption does not enter Table 1. It does limit any side comparison of unweighted multilevel coverage with Table E.2.

4.6 Appendix A variances

The question is which variance components generate the stated intraclass correlation. The answer is the pair printed in the Appendix A generator fragment.

The fragment fixes the scale that turns intraclass correlations .05 and .25 into a level-2 variance and a residual variance. We treated that fragment as the complete normal generator and did not estimate the variances from her tables. If the fragment we transcribed omits a scale factor, every normal bias would move by that factor. The Bernoulli threshold is on the probit scale and does not use those Gaussian variances.

4.7 Cluster-aligned random slope

The question is how the random slope enters the linear predictor. The answer is $u_{1j}x_{ij}$, with the slope draw aligned to cluster j.

WeMix then fits the weighted random-slope pseudo-likelihood under Method 2 scaling (Bailey et al., 2023). A slope vector that is not aligned to the cluster estimates a different covariance and a different level-1 coefficient. The match of the slope contrast to Table C.4 depends on this alignment.

5 Which ambiguities were resolved?

The question is which implementation choices were checked against an alternative and then fixed. The answer is the four checks below. Each one changes which number is allowed into the tables.

5.1 The two probit optimizers agree

The ambiguity was whether the Gauss–Hermite pseudo-likelihood had settled on a local mode. The check was a second optimization with BFGS, from the same start, on every 10th replication of the weighted multilevel probit. The finding is that the largest absolute gap on the latent intercept was under 0.0003 in both earlier Bernoulli files at intraclass correlation .05 (`simulation/results/bern_inf_05_5.csv`, `simulation/results/bern_inf_05_20.csv`). Those files were a 100-replication check of the optimizer. The reported factorial uses the same quadrature at 1,000 replications per cell. The implication is that the threshold contrast on that check is not an optimizer artifact. The reported means use `nlminb`.

5.2 Observation weights in glmmTMB do not match Table C.2

The ambiguity was whether a weighted probit mixed model with observation weights is the same estimator as the multilevel pseudo-likelihood. The check was glmmTMB with observation weights on the same Bernoulli samples, `https://CRAN.R-project.org/package=glmmTMB`. The finding is that the threshold mean at average cluster size 5 is about -0.974, against the printed true-weight mean of -0.170. At average cluster size 20 the glmmTMB threshold mean is about -0.247, against the printed -0.043. The implication is that this likelihood is a different estimator. It is not the contrast in Table 1, and it is not evidence against Table C.2.

5.3 GLMMadaptive weights are per cluster

The ambiguity was whether a mixed-model package that accepts a weight argument applies that weight once per observation. The check was the weight behavior of GLMMadaptive, `https://CRAN.R-project.org/package=GLMMadaptive`. The finding is that the package applies the weight once per cluster, not once per observation. The implication is that those fits are not the observation-level pseudo-likelihood behind Table C.2. They were not used as the reported Bernoulli means.

5.4 A recycled slope vector was discarded

The ambiguity was whether the random slope was indexed by cluster. The check was a first implementation that recycled a slope vector down the rows, followed by a refit in which the slope draw is constant inside the cluster, $u_{1j}x_{ij}$. The finding is that the recycled vector does not estimate the level-1 slope she printed. It was discarded. After the index was corrected, the approximate-weight minus true-weight slope contrast on 1,000 replications is 0.034 against the printed 0.033 (`simulation/results/cell_match.csv`). A separate 20-replication check compared a raw covariate with a centered covariate in the unweighted `lmer` fit. The reported random-coefficient results use the cluster-aligned slope.

6 How do the numbers compare?

The question is which printed numbers match, which are close, and which miss, once scope is separated from the third decimal. The answer is that 1,040 of 3,200 compared cells match, 1,531 are within one last printed place, and coverage matches in 90 of 752 cells. On the headline row, the slope contrast is close, the approximate-weight intercept contrast is close, and the normal intercept contrast, the Bernoulli threshold contrast, and the cluster-size-20 coverage are mismatches. Scope is limited on every row: the samples are a reconstruction in R, not Mplus 7, and not her seed stream.

Numerical agreement uses three labels at the printed precision. Match means our value rounds half up to the printed value. Close means it does not, and the absolute gap is at most one unit in the last printed place. Mismatch means the gap is larger than that. Scope is limited when the estimator and the scientific target are the ones she defined but the software or the seeds differ. The reported factorial uses her count of 1,000 replications. Limited scope does not change a numerical match. The page-1 tolerance of 0.001 is the sum of two half-thousandth rounding bounds, one for each bias that enters a contrast. That tolerance is a descriptive rule for Table 1. It is not an equivalence margin, and it is not the Monte Carlo standard error.

Table 2: Numerical agreement and reconstruction scope. Published values are taken from the dissertation tables named in the text. Our values are from this simulation. Precision is the printed unit used for the numerical label.

Comparison	Published	Ours	Precision	Numerical	Scope
Normal wtm minus ws intercept	0.151	0.152	0.001	mismatch	limited
Normal approx. minus true intercept	0.157	0.156	0.001	close	limited
Normal coverage contrast, cluster size 5	0.945	0.945	0.001	match	limited
Bernoulli wtm minus ws threshold	−0.170	−0.171	0.001	mismatch	limited
Random-coefficient slope contrast	0.033	0.034	0.001	close	limited
Coverage at cluster size 20	0.282	0.252	0.001	mismatch	limited
Raw normal wtm intercept bias	0.151	0.153	0.001	mismatch	limited
Non-informative intercept, worst signed cell	−0.002	0.002	0.001	mismatch	limited
Mean cluster count, primary cell	964	967.7	1	mismatch	limited

The normal intercept contrast is a mismatch because the unrounded gap is 0.001025, which is larger than 0.001, and 0.152 does not round to 0.151. The sign is the same. The approximate-weight intercept contrast is close: 0.156 does not round to 0.157, and the unrounded gap is 0.000764. The primary coverage contrast is a match at 0.945 against 0.945. The weighted single-level rate itself is 0.945 against the printed 0.946, which is close, and both sources show weighted multilevel coverage at essentially zero. The Bernoulli threshold contrast is a mismatch because the unrounded gap is −0.001098. The contrast stays negative, and the estimator order matches Table C.2. The random-coefficient slope contrast is close: 0.034 does not round to 0.033, and the unrounded gap is 0.000716. Coverage at average cluster size 20 and intraclass correlation .05 is a mismatch: 0.252 does not round to 0.282. The raw true-multilevel intercept bias on the primary normal cell is a mismatch, 0.153 against 0.151. The worst signed non-informative intercept is 0.002 against −0.002. The mean cluster count is a mismatch at the printed cluster and is within one percent, which is the agreement Table 4.1 supports for a Poisson design.

Table 3: Cell-level agreement on the finished factorial. A match rounds half up to the printed entry. Within one last place adds the cells whose unrounded gap is at most one printed unit. The 72 bracketed marginal root-mean-square-error cells are omitted.

Arm	Compared cells	Match	Within one last place
Normal random intercept	480	178	251
Bernoulli random intercept	672	228	341
Normal random coefficient	1,024	375	545
Bernoulli random coefficient	1,024	259	394
All compared cells	3,200	1,040	1,531
Coverage column	752	90	135

Of the 1,669 mismatches, 1,038 lie within two Monte Carlo standard errors of our mean. Bias matches or is within one last place in 518 of 824 cells. Mean square error does so in 669 of 824 cells. Relative bias matches in 50 of 592 cells. Cluster-specific root mean square error matches in 71 of 208 cells. The counts are in `simulation/results/match_summary.csv`.

The coverage gap of 0.030, from 0.282 to 0.252, is the unresolved cell in the next section. Both rates are far below the nominal 0.95, so the report that these intervals undercover still holds.

7 What remains unresolved?

The question is which published results this note does not establish. The answer is the coverage gap at average cluster size 20, the 72 bracketed marginal root-mean-square-error cells, and the appendix cells that miss the 0.001 rounding rule. All 208 settings now have about 1,000 replications.

Three candidates were open for that coverage gap: a documented finite-sample multiplier on the sandwich, ordinary Monte Carlo error, and an Mplus standard error that is not a scalar rescaling of the standard errors we saved. The saved CR-0 coverage on 984 replications is 0.252, against the printed 0.282 in Table E.2 (`simulation/results/coverage_scaling.csv`). Multiplying the variance, or the standard error, by $G/(G-1)$, by $N/(N-1)$, or by $G/(G-1) \times (N-1)/(N-3)$ moves the coverage from 0.252 to 0.253. That is about 0.001, not the gap of 0.030. The Monte Carlo standard error of our coverage is 0.014 (0.0138 before rounding to three decimals). A published rate of 0.282 on 1,000 draws has a binomial standard error of about 0.014 as well. The gap is about two of those standard errors, so Monte Carlo error can account for part of it and cannot show which standard-error definition produced 0.282. The largest remaining cause is an Mplus 7 MLR detail that the scalar factors do not reproduce. The artifact that would settle it is a Mplus 7 output for the informative normal cell at average cluster size 20 and intraclass correlation .05, ideally on the same replicates. The bias summary for that cell retains 965 replications (`simulation/results/summary.csv`); the scaling file retains 984 replications that have a saved standard error. Both coverages, 0.248 in the bias file and 0.252 in the scaling file, are far from 0.282 and far from 0.95.

The other open comparison is the bracketed marginal root mean square error. Those 72 cells, 24 random-intercept and 48 random-coefficient, are not in the checkpoints, which store the cluster-specific linear predictor. Of 3,200 compared cells, 1,040 match and 1,531 are within one last-place unit. Coverage is the weakest column: 90 of 752 coverage cells match. On the Bernoulli random-coefficient arm, the sandwich standard error for the incorrect-covariance weighted fits is finite in only a handful of replications in some cells, so those coverage rates use that smaller count. The point estimates in those cells still use about 1,000 replications. Mplus output would separate sandwich scaling from Monte Carlo error. No Mplus input file was found. The search is recorded in `notes/mplus-search.md`.

8 Where are the code and the source?

The question is whether a reader can obtain her programs and our results. The answer is that our result files are in this repository and her Mplus inputs were not found. The dated ledger is notes/availability.md, searched on 2 November 2026 (UTC). The Mplus search is notes/mplus-search.md, with the same date.

The dissertation PDF was recovered from the Internet Archive after the DigitalCommons download returned an access challenge. The copy used here is source/koziol2015.pdf. The title page matches Natalie A. Koziol, May 2015, with adviser James A. Bovaird. Appendix A prints part of the R generator. The dissertation names Mplus 7 for estimation and R with MASS for generation. It names no program archive, no OSF project, and no syntax file. The Figshare supplement of the 2017 article, file `hmbr_a_1292115_sm3492.docx` under https://doi.org/10.6084/m9.figshare.4743673, repeats simulation tables and does not contain Mplus or R programs. Searches of GitHub, the OSF API, Zenodo, the CYFS biography and CV, two later OSF preregistrations, and the public ORCID record did not produce the simulation inputs. The estimation programs were not obtained. The generator fragment in Appendix A was.

9 What should happen next?

The question is which artifact to obtain first, and which check to run on it. The answer is a Mplus 7 output for the informative normal cell at average cluster size 20 and intraclass correlation .05.

The check is a comparison of that output's intercept standard errors, and of the coverage they imply, with our coverage 0.252 and the printed coverage 0.282. If the output is computed on our replicates, the comparison isolates the sandwich. If it is computed on a new Mplus Monte Carlo, the comparison still shows whether Mplus 7 lands near 0.282 under the options on page 76. The same file would also show whether `WTSCALE=CLUSTER` matches the Method 2 scaling used here. The simulation that does not require Mplus is already the full factorial. What that run does not settle is the cluster-size-20 coverage gap, the 72 bracketed marginal root-mean-square-error cells, and the appendix cells that miss the printed thousandth. On some Bernoulli random-coefficient cells the sandwich standard error for the incorrect-covariance weighted fits is finite in only a small share of replications, so those coverage rates use that smaller count.

10 Compliance ledger

The question is whether the replication requirements that apply to this design were followed. The answer is that they were followed, with one departure: the preflight engine is for structural equation models, and these models are not structural equation models.

Table 4: Compliance ledger. A departure carries its justification in the evidence cell.

Requirement	Status	Evidence
Source identified and cited	followed	Natalie A. Koziol (2015), University of Nebraska–Lincoln, chair James A. Bovaird. DigitalCommons URL in the introduction.
Verdict before the design	followed	Abstract and Section 1 open with the replication verdict and the three contrasts.
Effect magnitude on page 1	followed	Table 1, with tolerance 0.001 and the point-estimate note.
Published numbers separated from ours	followed	Table captions name Tables C.2, C.4, E.2, and 4.1 as the source of published values.
Method citations	followed	Weighting, sandwich variance, pseudo-likelihood, and each package that was discussed are cited in the text.
Engine, lavaan / OpenMx	departed	Departure justification: the preflight expected lavaan and OpenMx. These models are not structural equation models, so that pair was not used to fit them. The weighted multilevel estimator was checked by WeMix 4.0.3 against the analytic pseudo-likelihood, and the two agree on the production replicate in `simulation/results/wemix_crosscheck_cell1_rep1.csv`.
Availability ledger	followed	`notes/availability.md` and `notes/mplus-search.md`, dated 2 November 2026.

A Artifacts

The question is which file holds each result, and what regenerates it. The answer is Table 5. Links point at the `master` copy of this repository and are navigation aids. The numbers in the note were read from the result files named here. The invocation, the working directory, and the seeds are recorded in `simulation/results/RUN.txt`. That file is the regeneration record. This appendix does not restate it as a second command.

Table 5: Files that hold the results in this note. Each link is the file on `master`.

Result	File	Regeneration record
Full-grid cell comparison, 3,272 printed cells	`simulation/results/cell_match.csv`	`simulation/compare_cells.py`, runners `simulation/run_grid_gaussian_rc.R`, `simulation/run_grid_bernoulli_ri.R`, `simulation/run_grid_bernoulli_rc.R`
Match rates by arm, statistic, and table	`simulation/results/match_summary.csv`	`simulation/compare_cells.py`
Published appendix cells	`simulation/results/published_cells.csv`	`simulation/parse_published.py`
Normal random-intercept replicates	`simulation/results/cell_01.csv`	`simulation/results/RUN.txt`, generator `simulation/generate.R`
WeMix cross-check, one replicate	`simulation/results/wemix_crosscheck_cell1_rep1.csv`	`simulation/results/RUN.txt`
Coverage scaling at cluster size 20	`simulation/results/coverage_scaling.csv`	`simulation/results/RUN.txt`
Dissertation PDF	`source/koziol2015.pdf`	Obtained from the Internet Archive; see `notes/availability.md`
Program search	`notes/mplus-search.md`	Dated ledger, 2 November 2026
Code and supplement ledger	`notes/availability.md`	Dated ledger, 2 November 2026
Bibliography source	`paper/references.bib`	Edited with the note

References

Tihomir Asparouhov. General multi-level modeling with sampling weights. *Communications in Statistics – Theory and Methods*, 35:439–460, 2006. doi: 10.1080/03610920500476598. Bibliographic details as printed in Koziol's reference list.

Paul Bailey, Blue Webb, Claire Kelley, Trang Nguyen, and Huade Huo. *WeMix: Weighted Mixed-Effects Models Using Multilevel Pseudo Maximum Likelihood Estimation*, 2023. URL https://CRAN.R-project.org/package=WeMix. R package version 4.0.3.

Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1):1–48, 2015. doi: 10.18637/jss.v067.i01.

David A. Binder. On the variances of asymptotically normal estimators from complex surveys. *International Statistical Review*, 51:279–292, 1983. Bibliographic details as printed in Koziol's reference list.

Paul Gilbert and Ravi Varadhan. *numDeriv: Accurate Numerical Derivatives*, 2019. URL https://CRAN.R-project.org/package=numDeriv. R package version 2016.8-1.1.

Søren Højsgaard, Ulrich Halekoh, and Jun Yan. The R package geepack for generalized estimating equations. *Journal of Statistical Software*, 15(2):1–11, 2006.

Milorad S. Kovačević and Shesh N. Rai. A pseudo maximum likelihood approach to multilevel modelling of survey data. *Communications in Statistics – Theory and Methods*, 32:113–135, 2003. Title and year as used by Koziol for the cluster-weight approximation. Not re-read against the journal.

Natalie A. Koziol. *A Comparison of Population-Averaged and Cluster-Specific Approaches in the Context of Unequal Probabilities of Selection*. Doctoral dissertation, University of Nebraska–Lincoln, 2015. URL https://digitalcommons.unl.edu/cehsdiss/229/.

Natalie A. Koziol, James A. Bovaird, and Sonia Suarez. A comparison of population-averaged and cluster-specific approaches in the context of unequal probabilities of selection. *Multivariate Behavioral Research*, 2017. doi: 10.1080/00273171.2017.1292115. DOI and author list taken from the Figshare record 10.6084/m9.figshare.4743673. The article PDF was not re-read for this note.

Thomas Lumley. Analysis of complex survey samples. *Journal of Statistical Software*, 9(1):1–19, 2004.

Thomas Lumley. *survey: Analysis of Complex Survey Samples*, 2024. URL https://CRAN.R-project.org/package=survey. R package version 4.2.1.

Linda K. Muthén and Bengt O. Muthén. *Mplus User's Guide*. Los Angeles, CA, 7 edition, 2012. Version named by Koziol on dissertation page 76.

Danny Pfeffermann, C. J. Skinner, D. J. Holmes, Harvey Goldstein, and J. Rasbash. Weighting for unequal selection probabilities in multilevel models. *Journal of the Royal Statistical Society, Series B*, 60:23–40, 1998. Bibliographic details as printed in Koziol's reference list.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024. URL https://www.R-project.org/. Version 4.3.3.

Sophia Rabe-Hesketh and Anders Skrondal. Multilevel modelling of complex survey data. *Journal of the Royal Statistical Society, Series A*, 169:805–827, 2006. Bibliographic details as printed in Koziol's reference list.